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The role of water vapour on the oxidation of two Ln–Si–Al–O–N glasses (Ln = Y, La)

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Abstract

The oxidation behaviour of LnSiAlON (Ln = Y, La) glasses was studied at different temperatures (990–1150 °C) and under different water vapour pressures (360–2690 Pa). These results were also compared with those obtained under O₂, N₂/H₂O or O₂/H₂O mixtures. When glasses are treated under a N₂/H₂O mixture, optical and SEM observations show porous scales. Transformations of the reaction rate data and a kinetic model show that there is only one limiting process occurring during oxidation. This rate limiting step is the progress of the chemical reaction at the internal interface. Determination of the pressure law dependence and thermodynamics calculations of water vapour molecules dissociation at the investigated temperatures allow us to suggest that the mechanism of oxidation corresponds to decomposition of water molecules on the oxynitride glass surface. © 2002 Elsevier Science B.V. All rights reserved.

1. Introduction

It is well known that oxidation resistance of silicon nitride-based ceramics is strongly influenced by the composition of the residual grain boundary phase [1–3]. Mixtures of different oxide additives (Y, La) have been widely used as sintering aids for such ceramics, usually resulting in a residual glassy phase. Jack [4] was the first to report the existence of oxynitride glasses in the Y–Si–Al–O–N system and subsequently considerable investigations were carried out to determine the extent of vitreous re-

gions in a wide range of M–Si–O–N and M–Si–Al–O–N systems [5–7]. The high temperature use of oxynitride glasses, in air, is limited by their oxidation resistance, but relatively few data are available in the literature. The oxidation rate of these Y, Ln systems varies with the nitrogen content, the nature and the concentration in lanthanide cations [8–10]. The comparison of the degradation behaviour of these glasses show that the presence of water vapour in the oxidising atmospheres has a significant effect on the oxidation kinetics [11,12].

In the present paper, oxidation kinetics of Y and La containing glasses are compared and oxidation mechanisms under water vapour are presented in order to define their use under isothermal conditions and to explain the degradation mechanisms of sintered materials at high temperature. The discussion of the results is focussed on the

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relationship between the oxidation rate constant and the partial pressure of water vapour.

2. Materials and procedures

2.1. Preparation and characterisation of glasses

Glasses were prepared using the melting procedure described by Hampshire et al. [13], from mixtures of 99.99% pure silicon nitride powder with high purity oxides (Al_2O_3 , Y_2O_3 , SiO_2 and La_2O_3). They were mixed in isopropanol, dried, and pressed into pellets.

Large batches (40–70 g) were then melted in a boron nitride lined graphite crucible under 100 kPa nitrogen at 1700 °C for 1 h. The glass was finally poured into a pre-heated graphite mould and annealed at 850–900 °C for 1 h and slowly cooled. The respective composition of the glasses are reported in Table 1.

The nitrogen content of the glass was obtained by elemental analysis using an analyser (Carlo Erba 1106) and by complete oxidation of the sample at 1150 °C, while silicon, aluminium and boron were analysed by ICP and oxygen content was given by infrared spectrometry of carbon monoxide formed after pyrolysis of the sample.²

Different analysis on various parts of the supplied glass bars showed a good homogeneity of the composition of the samples. Four hours XRD exposure time (diffractometer Siemens D5000) showed no presence of crystalline phase.

Some physical properties were determined before oxidation experiments (Table 2). Geometrical density is the average of about 15 measurements.

Thermal dilatometry (dilatometer Setaram TMA 92198, heating rate of 5 °C/min) under a flow of high purity nitrogen (N 50) enabled us to define the values of the glass transition temperature (T_g), the softening temperature (T_{ds}) and linear thermal expansion coefficient (α , in the tempera-

Table 1

Chemical composition of investigated glasses

	Cation	Si ⁴⁺	Al ³⁺	B ³⁺	O ²⁻	N ³⁻
Equiva-	La ³⁺	21.02	56.05	21.02	1.89	87.52
lent %						12.48
	Y ³⁺	18	47	35	–	80
Atomic	La ³⁺	9.15	18.30	9.15	0.82	57.14
%						5.43
	Y ³⁺	7.89	15.44	15.33	–	52.57
Weight	La ³⁺	41.93	41.93	8.14	0.29	31.16
%						2.51
	Y ³⁺	28.16	17.16	16.43	–	33.38
						4.87

Table 2

Physical properties of the glasses

	La glass	Y glass
ρ (g/cm ³)	3.85 ± 0.01	3.40 ± 0.01
T_g (°C)	891 ± 10	962 ± 10
T_{ds} (°C)	964 ± 10	1016 ± 10
$\alpha(10^{-6} \text{ K}^{-1})$	5.4	6 ± 0.4

ture range 200–800 °C). These results are an average on three different runs.

Glass samples for oxidation tests and dilatometry measurements were cut into cubes having an edge of around 4 mm and were polished on SiC paper grid (grade 1200, 2400 then 4000) before being cleaned in acetone in an ultrasonic bath, dried and weighted.

2.2. Oxidation experiments

Thermogravimetric analysis (Setaram B60) were conducted in an atmosphere controlled vertical alumina tube furnace with the sample suspended in an alumina crucible connected to the balance by a platinum chain. As this alumina tube and the crucible have been working for a long period, contamination of samples by alkali (Na and K) is negligible [14]. Samples were lowered into the hot zone of the furnace when the required temperature was reached and removed at the end of the run (isothermal conditions). Tests were performed under oxygen or mixtures: $P_{\text{O}_2} = 100$ kPa; $\text{N}_2/\text{H}_2\text{O}$ ($P_{\text{N}_2} + P_{\text{H}_2\text{O}} = 100$ kPa), $\text{O}_2/\text{H}_2\text{O}$ ($P_{\text{O}_2} + P_{\text{H}_2\text{O}} = 100$ kPa).

A known partial pressure of water vapour (360–2690 Pa) was introduced in the furnace by passing the inlet gas (high purity nitrogen N50), first,

² C.N.R.S.: Laboratoire Central d'Analyse, Vernaison (France).

through distilled water and then up a glass column, cooled or heated by ethylene glycol at a fixed temperature. Gas flow rate was maintained at 60 cm³/min to provide an excess of water vapour molecules to oxidise the samples. Characterisation of oxide layers and substrates were carried out by XRD (diffractometer Siemens D5000) and SEM (Philips XL 30) observations.

3. Oxidation results

During the oxidation of a SiAlON glass, nitrogen is replaced by oxygen to produce an oxide layer where 3 O²⁻ replace 2 N³⁻. The loss of nitrogen and gain of oxygen leads to a net weight gain in the sample. This is used to define the degree of advancement of the reaction at any given time, α , as follows:

$$\alpha = \frac{\Delta m_t}{\Delta m_\infty}, \quad (1)$$

where Δm_t is the weight gain at time t and Δm_∞ is the weight gain after an infinite time, when all nitrogen is replaced by oxygen. In this case Δm_∞ corresponds to 1.66 and 3.48 wt%, respectively for La and Y containing glass.

3.1. Oxidation of the glasses under water vapour

Oxidation tests for lanthanum glasses were conducted at different temperatures (990–1050 °C) under 2100 Pa (Fig. 1(a)) and at 1000 °C under different water vapour pressures (400–2340 Pa) (Fig. 1(b)). In the same way, oxidation runs for yttrium glasses were conducted in the temperature range from 1050 to 1155 °C under 2100 Pa (Fig. 2(a)) and at 1150 °C under different water vapour pressures (360–2690 Pa) (Fig. 2(b)). Curve plots are more regular for this glass. Water vapour mixed with high purity nitrogen (N50) is an effective oxidant as illustrated by Figs. 1 and 2. For the same range of temperature, oxidation rate is faster than in dry oxygen [12,15]. An advancement of 0.3 is reached after 13 h at 1035 °C for lanthanum glass, and after 7 h at 1105 °C for yttrium glass, both under 2100 Pa water vapour pressure. As reported in a previous work, the oxidation behaviour of LnSiAlONs glasses depends on the

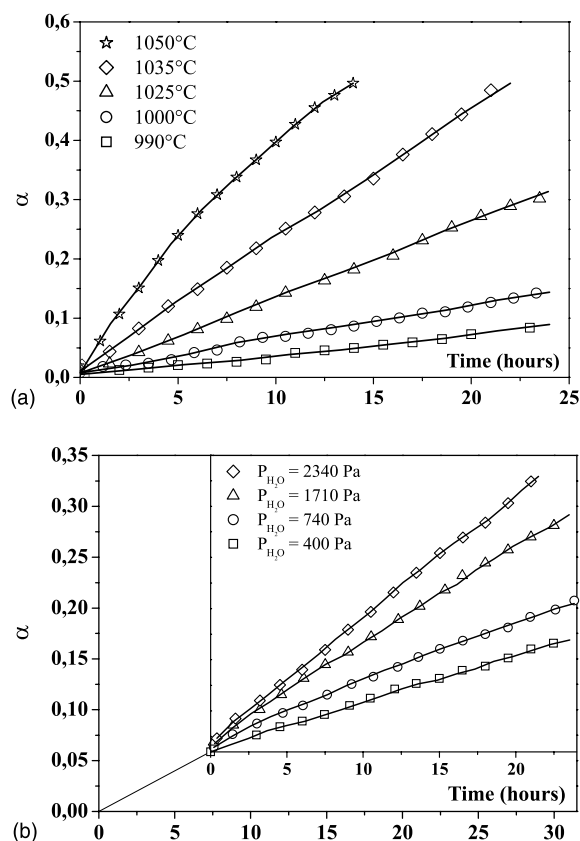


Fig. 1. Advancement versus time for lanthanum glass: (a) Under nitrogen + 2100 Pa water vapour pressure and (b) under nitrogen plus different water vapour pressures at 1000 °C, after a pre-oxidation of 7.5 h at 1000 °C under a partial water vapour of 1200 Pa.

nature of the dopant cation and the contents of nitrogen or of the rare earth cation [10]. It means that the reactivity is related to the compactness and the structure of the glass.

At low temperature, the advancement for lanthanum glass is almost linear with time. A pre-oxidation run has been done at 1000 °C during 7.5 h, in order to have an initial advancement α of 0.06 to verify if there is no change in the oxidation mechanism (fixed interface area). Advancement of the reaction is always linear with time, in the studied interval, for all the samples as the water vapour pressure varies (Fig. 1(b)). No latent period is observed.

Optical and SEM observations indicate that the scale is porous. The reaction is inclined to take

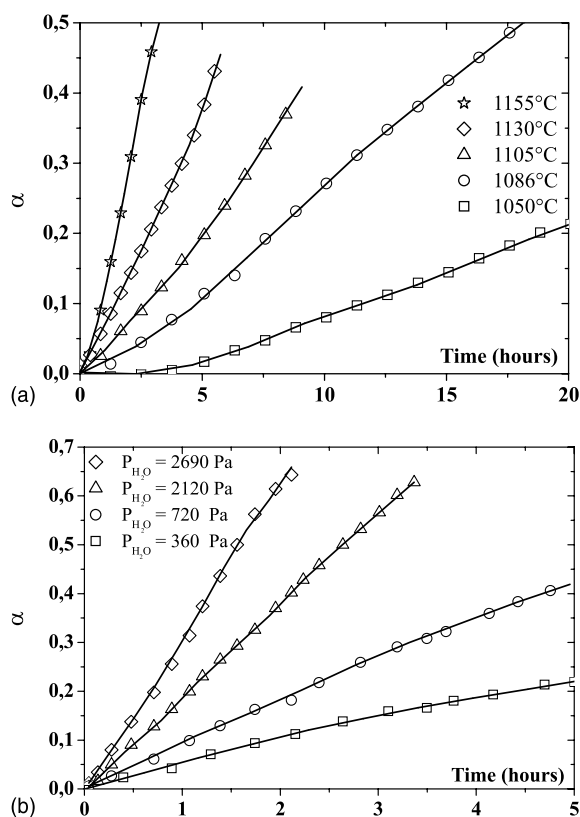


Fig. 2. Advancement versus time for yttrium glass: (a) Under nitrogen + 2100 Pa water vapour pressure and (b) under nitrogen plus different water vapour pressures at 1150 °C.

place preferentially at the centre of the edges (Fig. 3(a)). Gas bubbles are formed at the internal interface during oxidation and are responsible for the porosity. This also means that the scale has a plastic behaviour or is viscous at the investigated temperature. A back-scattered image (Fig. 3(b)) highlights that there is no change in composition of heavy elements in the glass, as also confirmed by EDX profiles through the surface of the oxidised sample.

No crystallisation of the substrate can be detected up to 1050 °C. XRD analysis of the scale shows the formation at all temperatures of W oxide phase $\text{La}_4\text{Si}_8\text{Al}_6\text{O}_{31}$ [16] and lanthanum disilicate (JCPDS 44-0346) as the main reaction products.

Oxide scale is also porous and expanded in the temperature range studied for yttrium glass, as illustrated by Fig. 4. The oxide scale comprises mostly α , Y (keiviite), and β yttrium disilicate and

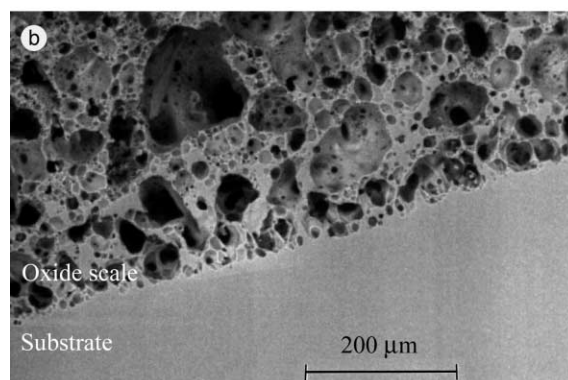
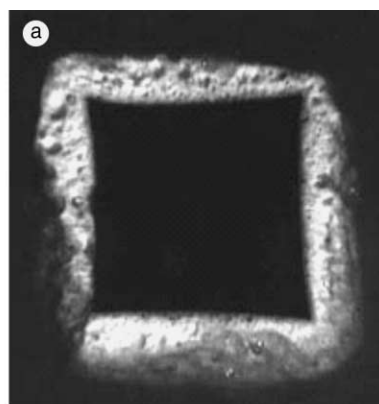


Fig. 3. Optical (a) and SEM backscattered (b) observations of lanthanum glass oxidised at 990 °C under 400 Pa water vapour pressure.

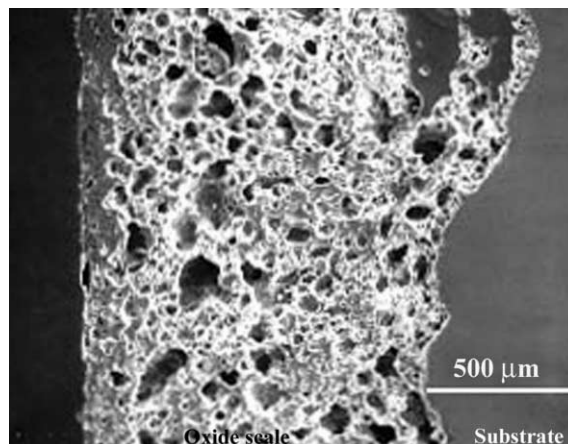


Fig. 4. Oxide scale of yttrium glass.

mullite. For both glasses, pores and the porous volume are more important after oxidation under water vapour, than after oxidation under dry oxygen.

3.2. Analysis of the results

Two approaches are used to discuss the results and to compare the oxidation resistance of the glasses under water vapour: the affine transformation of the advancement curves and a kinetic modelling.

3.2.1. Affine transformation

From the shape of the curves of Figs. 1 and 2, an affine transformation can be applied. This is a two step mathematical processing of the data

made without any assumption about the reaction mechanisms [17].

In a first step, a shift in the time scale for all curves, except for one set as the reference, is applied. The ratio A is calculated using the following equation:

$$\frac{t_{T,P(\text{ref})}}{t_{T,P}} \bigg|_{\alpha} = A, \quad (2)$$

where A is the average affinity coefficient, $t_{T,P(\text{ref})}$ is the time for a fixed advancement of the reference temperature or pressure and $t_{T,P}$ is the time for an isothermal or isobaric curve for a fixed advancement. Figs. 5 and 6 illustrate the advancement in function of the reduced times ($t_{T,P} \times A$) for respectively lanthanum and yttrium glasses.

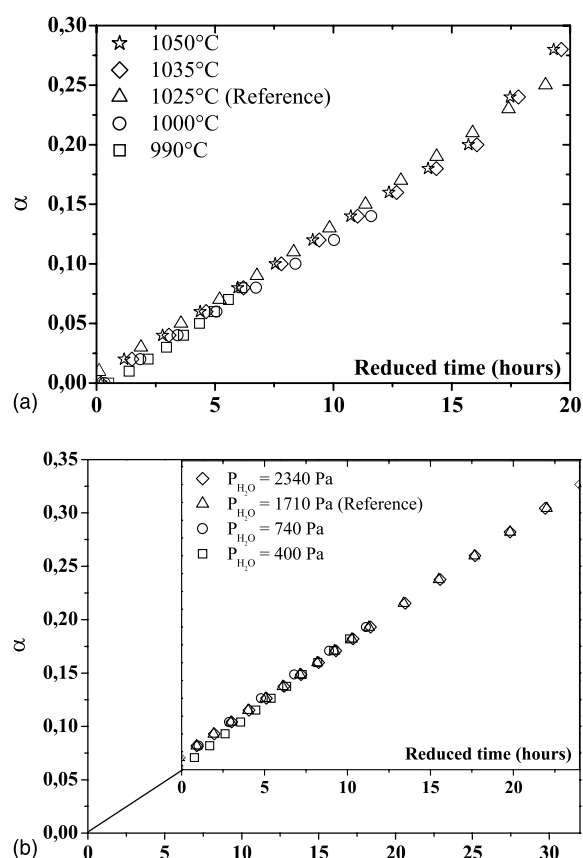


Fig. 5. Affine transformation for lanthanum glass, influence of (a) temperature under $N_2 + 2100$ Pa H_2O vapour pressure and (b) water vapour pressure at 1000°C .

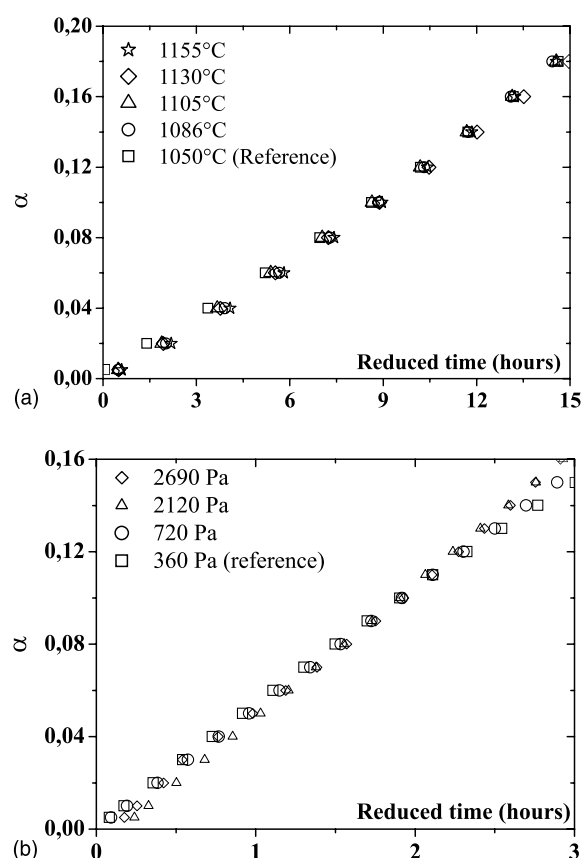


Fig. 6. Affine transformation for yttrium glass, influence of (a) temperature under $N_2 + 2100$ Pa H_2O vapour pressure and (b) water vapour pressure at 1150°C .

In the second step of the affine transformation a calculation of the derivative of the advancement versus time ($d\alpha/dt$) has to be carried out and, at each fixed value of α , the following equation has to be verified:

$$\left. \frac{\left(\frac{d\alpha}{dt}\right)_{T,P}}{\left(\frac{d\alpha}{dt}\right)_{T,P(\text{ref})}} \right|_{\alpha} = A', \quad (3)$$

where A' is the average coefficient of affinity, $(d\alpha/dt)_{T,P}$ is the rate of advancement for an isothermal or isobaric curve $(d\alpha/dt)_{T,P(\text{ref})}$ and is the rate of advancement of the reference temperature or pressure [17].

Figs. 7 and 8 show the rate of advancement ($(d\alpha/dt)_{T,P} \times A'$) in function of the advancement for lanthanum and yttrium glass, respectively.

When all the values of α or $(d\alpha/dt)$ are on the same line, it indicates that there is only one process

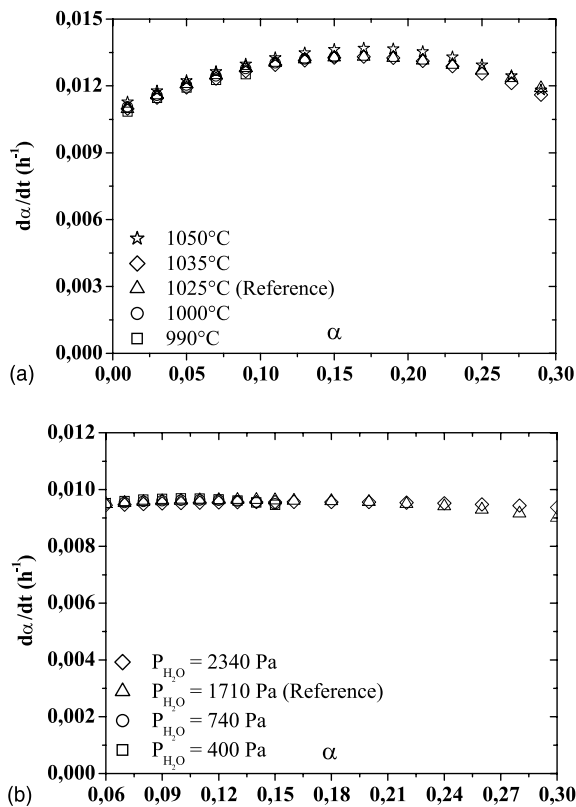


Fig. 7. Affine transformation for lanthanum glass, rate of advancement in function of the advancement, influence of (a) temperature under $N_2 + 2100$ Pa H_2O vapour pressure and (b) water vapour pressure at $1000^\circ C$.

of oxidation occurring in this investigated range of temperature or pressure, as it is verified for the oxidation of La and Y glasses.

3.2.2. Kinetic modelling

The shape of the advancement curves with respect to time and the porosity of the scale, which is formed at the internal interface, for both glasses, suggest that the limiting step is the chemical reaction at the internal interface. When the limiting step is the chemical reaction at the internal interface, and for cubic samples, a shrinkage cube model can be used:

$$F(\alpha) = 1 - (1 - \alpha)^{1/3} = \frac{2kt}{a_0} = kt, \quad (4)$$

where k is the rate of the reaction and a_0 is the initial side of the cube [17].

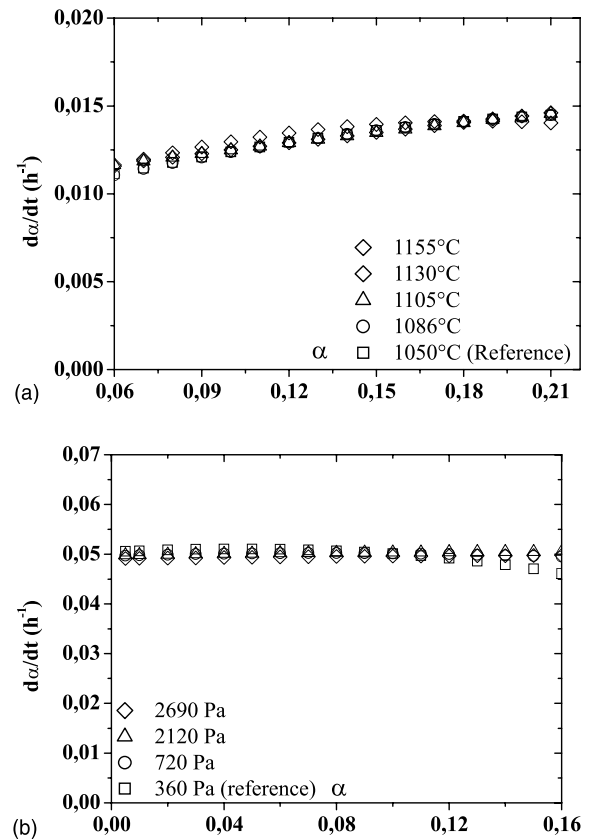


Fig. 8. Affine transformation for yttrium glass, rate of advancement in function of the advancement, influence of (a) temperature under $N_2 + 2100$ Pa H_2O vapour pressure and (b) water vapour pressure at $1150^\circ C$.

The linear nature of the function $F(\alpha)$ for the investigated temperature ranges confirms the use of Eq. (4) for these glasses (Figs. 9 and 10).

3.2.3. Determination of the apparent activation energy and the power law exponent

The standard deviation of all the calculated results lies within a confidence interval of ± 1 .

From affine transformation results, an apparent activation energy (E) can be determined, by plotting $\ln(A)$ as a function of $1/T$, where A is the average coefficient of affinity (Fig. 11) [17]. Values of $590 \pm 10 \text{ kJ mol}^{-1}$ for lanthanum glass and $370 \pm 10 \text{ kJ mol}^{-1}$ for yttrium glass were found.

From the slope of $F(\alpha)$ in function of time, the rate constant of the oxidation reaction, k , is de-

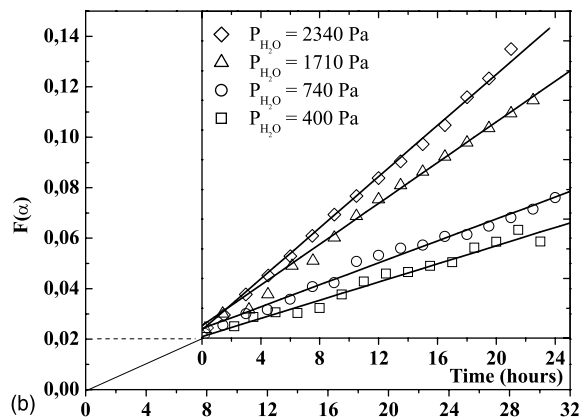
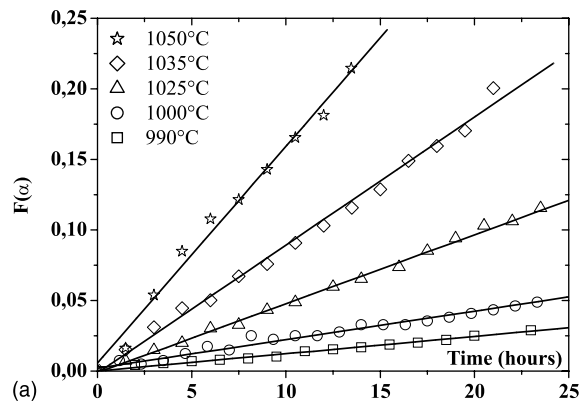


Fig. 9. $F(\alpha)$ for lanthanum glass: (a) Under nitrogen + 2100 Pa water vapour pressure and (b) under nitrogen plus different water vapour pressures at 1000 °C after preoxidation of 7.5 h at 1000 °C, under a partial water vapour pressure of 1200 Pa.

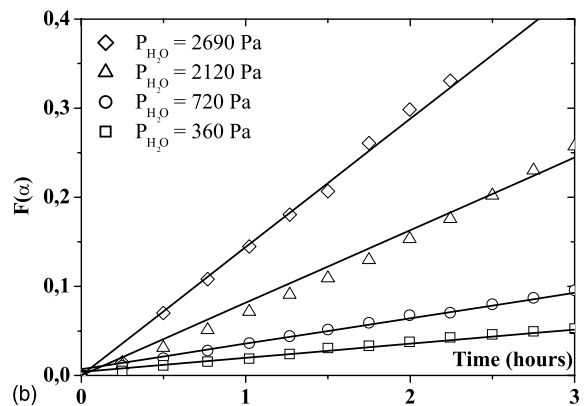
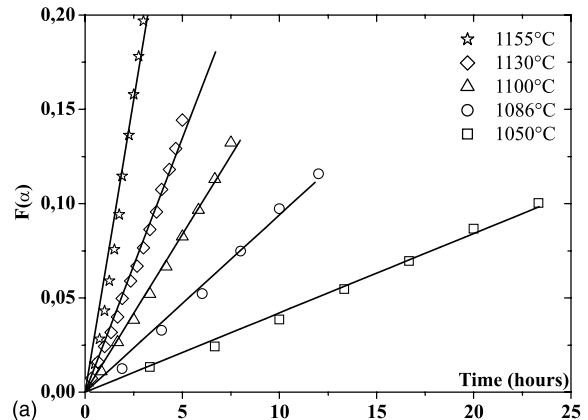


Fig. 10. $F(\alpha)$ for yttrium glass: (a) Under nitrogen + 2100 Pa water vapour pressure and (b) under nitrogen plus different water vapour pressures at 1150 °C.

termined. By plotting $\ln(k)$ versus $1/T$ (Fig. 12), an apparent activation energy (E) can be calculated when the Arrhenius law is verified and is found equal to $580 \pm 10 \text{ kJ mol}^{-1}$ for lanthanum glass and $410 \pm 10 \text{ kJ mol}^{-1}$ for yttrium glass. These results are in good agreement with those obtained by the affine transformation.

By using the affine transformation, the effect of water vapour on the oxidation rate, V , was calculated using the following equation: $V = k'P_{\text{H}_2\text{O}}^n$ [17]. The n parameter was then estimated by plotting $\ln(A)$ as a function of $\ln(P_{\text{H}_2\text{O}})$ where A is the average coefficient of affinity (Fig. 13) [17]. The n exponent was found to be equal to 0.61 ± 0.03 and 0.74 ± 0.05 for lanthanum and yttrium glass, respectively.

It is also possible to determine the effect of the pressure on the oxidation rate of all the samples

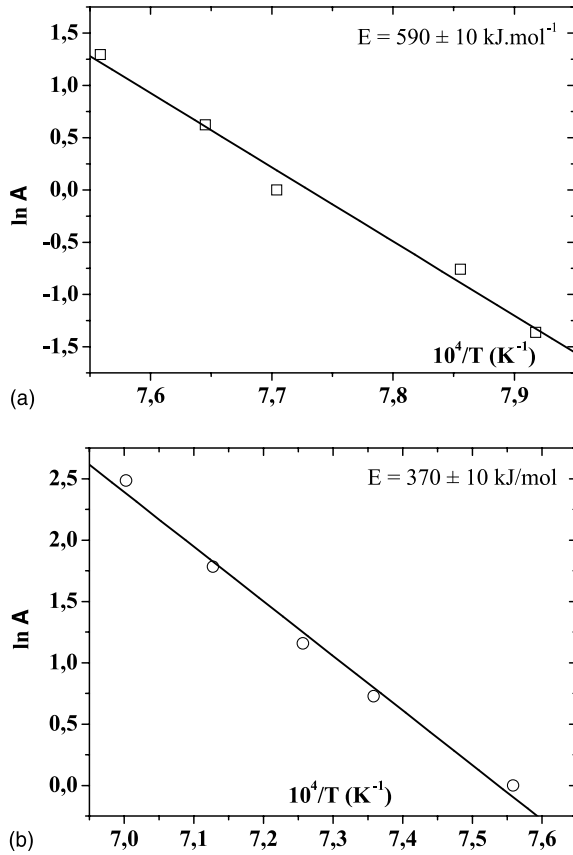


Fig. 11. Activation energy for oxidation under 2100 Pa water vapour pressure, using the affine transformation: (a) lanthanum glass and (b) yttrium glass.

from the plot of $\ln(k')$ as a function of $\ln(P_{H_2O})$ (Fig. 14). The value of the n exponent is deduced from the results of Figs. 9(b) and 10(b).

According to the results of these fits, the pressure exponent for lanthanum samples is $n = 0.62 \pm 0.03$, which is very close to the values found for yttrium glass $n = 0.68 \pm 0.07$ in the same range of water vapour pressures. Moreover, one notices that the n exponent calculated at the beginning of the reaction ($\alpha = 0$) or during the oxidation ($\alpha = 0.06$) are similar which means that the presence of an oxide scale has no influence on the limiting step (Fig. 9(a) and (b)).

Table 3 summarizes all the calculated values of the apparent activation energy and of the power law exponent, estimated by the different methods.

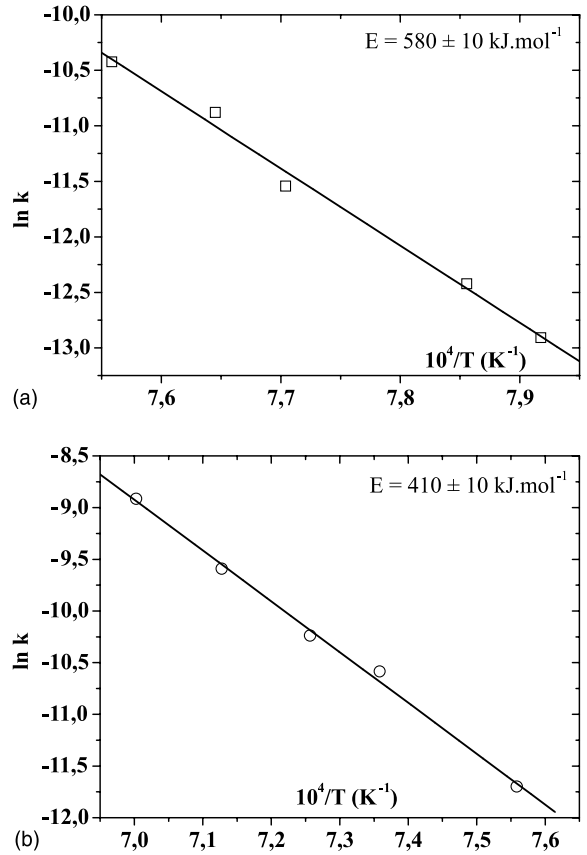


Fig. 12. Arrhenius plot for oxidation under 2100 Pa water vapour (a) lanthanum glass and (b) yttrium glass.

Because the curves of the advancement versus time are affine (Figs. 5–8), the rate of oxidation, V , can be expressed by an equation, with separated variables, as follows [17]:

$$V = A(\alpha)B(T)C(P), \quad (5)$$

where $A(\alpha)$ represents the reaction area for a cubic symmetry, as $A(\alpha)$ is a function of the fractional weight gain, $B(T)$ and $C(P)$ the influences of the temperature and the pressure respectively. Thus, the kinetic equations are, according to kinetic model results, for lanthanum glass:

$$V = a(1 - \alpha)^{2/3} P^{0.62} \exp\left(\frac{-580 \times 10^3}{RT}\right), \quad (6)$$

and for yttrium glass:

$$V = b(1 - \alpha)^{2/3} P^{0.68} \exp\left(\frac{-410 \times 10^3}{RT}\right). \quad (7)$$

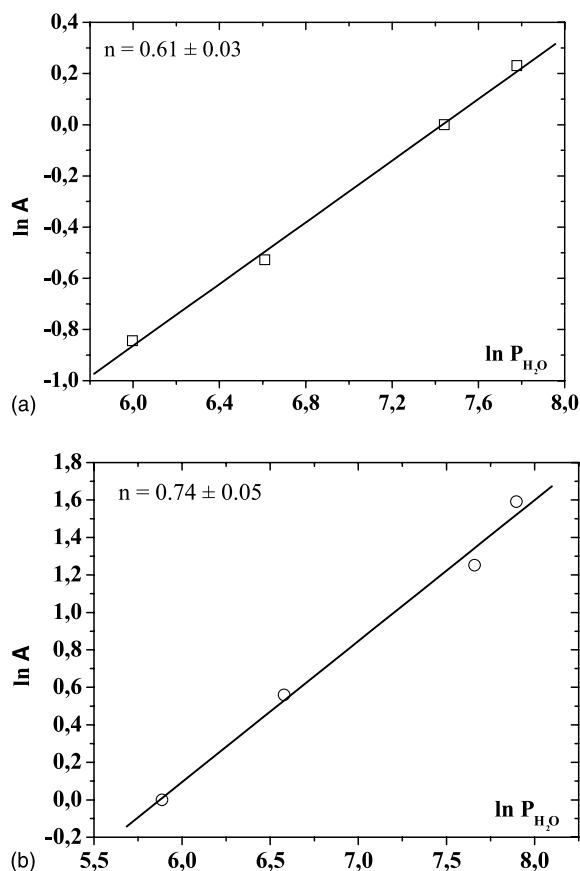


Fig. 13. Determination of the pressure exponent by an affine method: (a) lanthanum glass and (b) yttrium glass.

3.3. Discussion

Under a water vapour atmosphere, different species can react with oxynitride glasses such as molecular water or other species due to the decomposition of water at high temperature. In order to determine the concentration of these species, thermodynamics calculations were carried out as reported on Table 4 at 1000 and 1150 °C [18]. The software used is based on the minimisation of the free enthalpy according to Lagrange, White and Dantzig algorithms [18]. Influence of the water vapour pressure is expressed in number of moles of H_2O with respect to N_2 in Table 4. According to these calculations, dissociation of molecular water by temperature is low in our experimental conditions. The possible reactions and the concentration

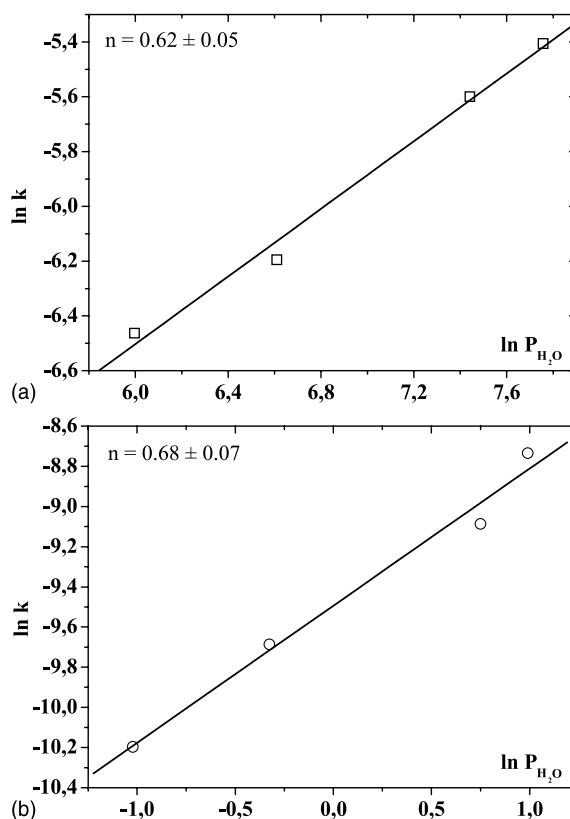


Fig. 14. Determination of the pressure exponent by the kinetics method: (a) lanthanum glass and (b) yttrium glass.

Table 3

Summary of the calculated values of enthalpy of reaction and n exponent by the different methods

Method	Glass	ΔH_{rxn} (kJ mol ⁻¹)	n
Affine transformation	La	590 ± 10	0.61 ± 0.03
	Y	370 ± 10	0.74 ± 0.05
Kinetic model	La	580 ± 10	0.62 ± 0.05
	Y	410 ± 10	0.68 ± 0.07

of the various oxidising species determined from the corresponding equilibria are reported in Table 5.

In order to compare the reactivity of the oxidising species, tests have been performed under dry oxygen, a mixture of nitrogen plus water vapour or a mixture O_2/H_2O , as illustrated by Fig. 15. The

Table 4

Decomposition of water in the used experimental conditions

Initial number of moles	Nb of moles at	H ₂ O	OH'	H•	H ₂	O ₂
N ₂ : 100; H ₂ O: 2	1050 °C and 2340 Pa H ₂ O pressure	4	6.37×10^{-5}	10^{-6}	8.17×10^{-4}	1.79×10^{-4}
N ₂ : 1000; H ₂ O: 4	1050 °C and 400 Pa H ₂ O pressure	2	1.95×10^{-5}	1.63×10^{-7}	2.16×10^{-4}	6.36×10^{-5}
N ₂ : 100; H ₂ O: 2	1150 °C and 2340 Pa H ₂ O pressure	4	3.12×10^{-4}	1.12×10^{-5}	3.06×10^{-3}	6.34×10^{-4}
N ₂ : 1000; H ₂ O: 4	1150 °C and 400 Pa H ₂ O pressure	2	9.59×10^{-5}	1.81×10^{-6}	8.07×10^{-4}	2.27×10^{-4}

Table 5

Water vapour pressure dependence for various oxidising species

Decomposition reaction for water vapour	$P_{\text{H}_2\text{O}}$ dependence of oxidising agents
No decomposition	$[\text{H}_2\text{O}] = K_1(P_{\text{H}_2\text{O}})^1$ (1)
$\text{H}_2\text{O} \leftrightarrow \text{H}^\bullet + \text{OH}'$	$[\text{OH}'] = K_2(P_{\text{H}_2\text{O}})^{1/2}$ (2)
$\text{H}_2\text{O} \leftrightarrow \text{H}_2 + \frac{1}{2}\text{O}_2$	$[\text{O}_2] = K_3(P_{\text{H}_2\text{O}})^2/(P_{\text{H}_2})^2$ (3)

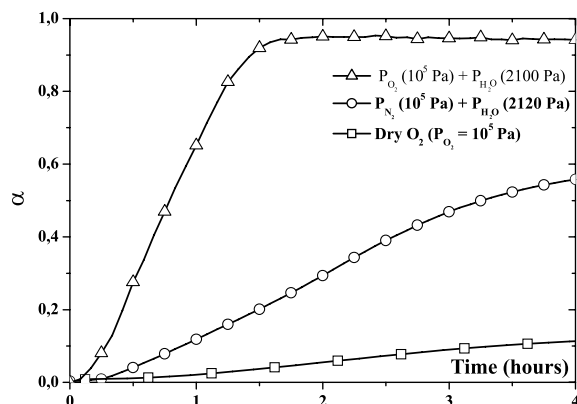


Fig. 15. Comparison of advancement under dry oxygen, mixture of N₂/H₂O and a mixture of O₂/H₂O for yttrium glass at 1150 °C.

rate of oxidation of the oxynitride glass, at a given time and for a low water vapour pressure, is higher than in dry oxygen. In the case of a O₂/H₂O mixture the oxidation rate is higher than the sum of the rates of the separate oxidizing gases. This demonstrates that a synergy takes place between the action of the two gases. Moreover, the fact that water vapour strongly enhances the oxidation indicates that water vapour reacts preferentially with

the oxynitride glass compared to molecular oxygen. This means that the third reaction for the decomposition of water in Table 5 can be ruled out when we take into account the very low concentration of oxygen. SEM observations show that the size of the pores in the oxide scale is higher in a water vapour atmosphere than with oxygen. This higher porous volume, due to lower viscosity of the oxide scale in presence of water vapour [19,20], facilitates the access of the oxidizing species to the internal interface. If reaction 1 in Table 5 is verified, an oxidation with a limiting step corresponding to a progress of the internal interface implies that the concentration of water at the internal interface is proportional to $(P_{\text{H}_2\text{O}})^1$ and the pressure exponent is equal to 1 ($n = 1$). In the case of the second reaction, the reaction rate with OH' is proportional to $(P_{\text{H}_2\text{O}})^{1/2}$. The pressure law, determined for both oxynitride glasses, gives an exponent ranging from 0.60 to 0.70, which indicates that the mechanism of oxidation corresponds to decomposition of water molecules on glass surfaces, according to reaction two in Table 5. Structural studies of oxynitride glasses have shown that nitrogen is mainly bonded to silicon [21], thus, the reaction with water vapour could be described by an hydrolysis of Si–N bonds.

4. Conclusion

The role of water vapour on the oxidation of two LnSiAlON glasses (Ln = La, Y) has been studied. Experiments were carried out under O₂/N₂, N₂/H₂O or O₂/H₂O mixtures in the tempera-

ture range 990–1200 °C. It was found that water vapour acts as an effective oxidant. At a low partial pressure ($P_{\text{H}_2\text{O}} = 2120$ Pa), water vapour in nitrogen gives a significantly higher advancement of reaction than an atmosphere of pure dry oxygen. The analysis of the results, according to different methods (affine transformations, kinetics law), show that the reaction of water vapour with the oxynitride glass at the internal interface (oxide scale/oxynitride substrate) is the limiting step.

The pressure exponent determined from the rates of reaction allow us to suggest a mechanism of oxidation which corresponds to decomposition of water molecules on glass surface.

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